

Shubham Kaushik

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RESEARCH INTERESTS

Databases, Data systems, Storage systems, Distributed systems, Data streaming, Cloud Data Platforms

PROFESSIONAL EXPERIENCE

Jan 2024 - Present	Ph.D. Researcher Brandeis University, MA, United States
Mar 2022 - Aug 2022	Software Engineer , Scalability & Infrastructure Kwalee, India
Jun 2021 - Mar 2022	Engineer - Information Security , Information Security Automation FIS Global, India
Oct 2019 - Jun 2021	Project Engineer , Cloud Computing Wipro Limited, India
Jul 2018 - Oct 2019	Project Engineer , Big Data Security Wipro Limited, India
Mar 2017 - Apr 2017	Full Stack Developer Intern , Full Stack Development SoPo Internet Private Limited, India

EDUCATION

Jan 2024 - Present	Doctor of Philosophy (Ph.D.) GPA: 3.98/4.0 Brandeis University, MA, United States Major: Computer Science
Sep 2022 - Dec 2023	Master of Science (M.S.) GPA: 3.88/4.0 Boston University, MA, United States Major: Computer Science with specialization in "Data-Centric Computing"
Jul 2014 - Jun 2018	Bachelor of Technology (B.Tech.) Maharshi Dayanand University, Haryana, India Major: Computer Science & Engineering Thesis: "Fault Modelling of an Object-Oriented System using Colored Petri Nets"

PUBLICATIONS

In Review	Boao Chen, Shubham Kaushik , and Subhadeep Sarkar. <i>Mind the Buffer: Dissecting the LSM-Buffer Design Space</i> [Read Me]
VLDB 2026	Abhishek Chanda, Shubham Kaushik , Artem Lavrov, and Subhadeep Sarkar. <i>TexBench: A Unified Benchmarking Suite for Shifting Workloads</i> , 52nd International Conference on Very Large Data Bases
ICDE 2026	Shubham Kaushik , Manos Athanassoulis, and Subhadeep Sarkar. <i>RangeReduce: Query-Driven Compaction In LSM-Trees</i> , 42nd IEEE International Conference on Data Engineering [Talk]
TPCTC 2025	Alexander H. Ott, Shubham Kaushik , James Chen, and Subhadeep Sarkar. <i>Tectonic: Bridging Synthetic and Real-World Workloads for Key-Value Benchmarking</i> , TPC Technology Conference on Performance Evaluation & Benchmarking
DBTest 2024	Shubham Kaushik and Subhadeep Sarkar. <i>Anatomy of the LSM Memory Buffer: Insights & Implications</i> , In Proceedings of the International Workshop on Testing Database Systems
JCSE 2019	Shubham Kaushik and Ratneshwer. <i>Fault Modeling of an Object-Oriented System using CPN</i> , International Journal of Computer Sciences and Engineering

POSTERS

NES Day 2026	Alexander H. Ott, Shubham Kaushik , Boao Chen, and Subhadeep Sarkar <i>Tectonic: Bridging Synthetic and Real-World Workloads for Key-Value Benchmarking</i> , 2026 New England Systems Day
NEDB Day 2025/2024	Shubham Kaushik , Manos Athanassoulis, and Subhadeep Sarkar <i>RangeReduce: A Range Query Driven Compaction for LSM-Trees</i> , North East Database Day

TALKS

05/2026	"RangeReduce: Query-Driven LSM Compactions", ICDE'26 , Montréal, Canada
04/2026	"RangeReduce: Query-Driven LSM Compactions", MidAS Seminar , Boston University, USA
01/2026	"Tectonic: Bridging Synthetic and Real-World Workloads", 2026 New England Systems Day , Harvard, USA
10/2025	"Range Queries and LSM-Trees: From Foes to Friends", Salesforce , USA
09/2025	"Tectonic: Bridging Synthetic and Real-World Workloads for Key-Value Benchmarking", TPCTC , UK
06/2024	"Anatomy of LSM Memory Buffers: Insights and Implications", DBTest , Santiago, Chile

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Java, Python, SQL, Rust, Golang
- **Markup Languages:** HTML, CSS, JSON, YAML, $\LaTeX$, Markdown
- **Databases:** RocksDB, Postgres, Redis, SQLite, ORM
- **Tools & Systems:** Linux, Asyncio, Git, Docker, Django, Django-REST, AWS, Apache Flink

KEY PROJECTS

- **FluidLSM** (*Ongoing*): LSM-trees offer excellent write throughput but their performance is highly sensitive to configuration, which is a huge design space. Real workloads shift over time, and no static configuration remains optimal. FluidLSM adaptively reconfigure the LSM-tree on the fly, switching compaction strategies and tuning parameters in response to workload shifts, without manual intervention. [[Blog](#)]
- **Tectonic+** (*Ongoing*): A next-generation benchmarking suite designed to replace YCSB and other static workload generators for key-value stores. Tectonic+ extends the idea of Tectonic while adding multiple storage backends (RocksDB, Cassandra, Redis, ScyllaDB) through a unified database abstraction layer. [[Blog](#)] [[GitHub](#)]
- **Mind the Buffer** (*Under Review*): A comprehensive study of the LSM in-memory buffer design space, which is a small yet critical component of LSM-tree through which every read and write must pass. Evaluate nine buffer implementations across 1,200+ experiments, and present a practitioner handbook with 10 guidelines for buffer selection. Also, introduce an Adaptive buffer that monitors workload composition and switches implementations at flush boundaries, consistently matching the best under shifting workloads. [[Blog](#)] [[GitHub](#)]
- **RangeReduce** (*Published, ICDE 2026*): LSM trees suffer from poor range query performance due to high read amplification and redundant merging across sorted runs. RangeReduce piggybacks on range queries to compact data already read into memory, writing back only valid entries at a higher level — reducing read amplification and compaction overhead without adding a separate compaction pass. It achieves up to 18% lower range-query latency, 90% reduction in compaction debt, 12% less data movement, and 20% lower space utilization. [[Paper PDF](#)] [[Blog](#)] [[GitHub](#)]
- **Tectonic** (*Published, TPCTC 2025*): A Rust-based, highly configurable key-value workload generator that models the temporal, structural, and dynamic properties of real-world workloads, properties that YCSB and others cannot capture. Tectonic supports composite key generation, dynamic workload shifts, and user-specified data sortedness, achieving 2× higher throughput and 84% lower memory footprint than the state-of-the-art. [[Paper PDF](#)] [[Blog](#)] [[GitHub](#)]

TEACHING ASSISTANT

Spring 2026/25/24 | *Database Management Systems (COSI 127B)*, Brandeis University [[SP '26](#)] [[SP '25](#)] [[SP '24](#)]
Fall 2024/2025 | *Introduction to Computer Networking (COSI 128A)*, Brandeis University
Fall/Spring 2023 | *Data Mechanics (DS 310)*, Boston University
Fall 2022 | *Computer Networks (CS 455)*, Boston University

PROFESSIONAL SERVICES

2026 | Program Committee Member, VLDB 2027.
2026 | Member, VLDB 2026 Shadow PC.
2025 | Web Chair for *Northeast Database (NEDB) Day 2025*.
2024/2025 | External reviewer for IEEE International Conference on Big Data.
2024 - Present | Member, Association for Computing Machinery (ACM).
2024 - Present | Student Member, Institute of Electrical and Electronics Engineers (IEEE).

CERTIFICATION

Jul 2023 | “*The Ultimate Hands-On Hadoop: Tame your Big Data!*” - Udemy [[link](#)]
Jul 2023 | “*Beginning C++ programming from Beginner to Beyond*” - Udemy [[link](#)]
Oct 2018 | Statement of accomplishment for “*Python Track*” - DataCamp [[link](#)]

EXTRA CURRICULAR ACTIVITIES

Sep 2023/2025 | Judged and mentored at *HackMIT 2023*, aiding teams with technical challenges.
Nov 2022 | Mentored 4 teams, with an average of 20 participants at *BostonHacks*.
Jan 2017 | Volunteered in the Program Event Management team at the *National Youth Festival*.